

PaymentProcessor.java - Explanation

ALGORITHM / EXECUTION FLOW:

1. addPayment() puts a payment into the generic queue.
2. processAll() loops through the queue.
3. For each item, it calls payment.process().
4. The JVM uses Dynamic Dispatch to figure out if it's a CreditCard, PayPal, or Bank object, and runs that specific code.
5. It saves the results to history and clears the queue.

SECTION 1: THE DATA STRUCTURES

Notice that the list type is `Payment`. It is NOT `CreditCardPayment`. Because of inheritance, a List of Payments can hold ANY child class.

SECTION 3: THE POLYMORPHIC ENGINE

DYNAMIC DISPATCH HAPPENS HERE: We do NOT write 'if (payment instanceof CreditCard)'. We just call process(). Java automatically looks at the object in memory, sees its true identity, and runs the correct override.